Discrepancy and Duality

1. Discrepancy
   - a problem in computer graphics
   - computing the discrepancy
   - computing extrema
2. Duality
   - duality transforms
   - the dual of the discrepancy problem
3. An Example with CGAL
   - finding collinear points via duality

MCS 481 Lecture 24
Computational Geometry
Jan Verschelde, 19 October 2026

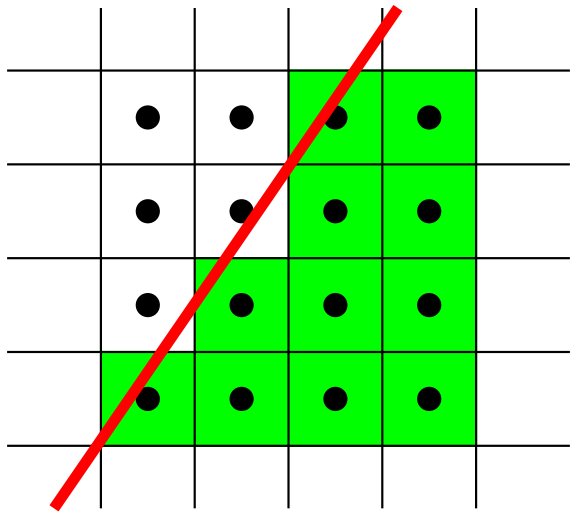
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rendering images

Ray tracing is a technique to determine the visible object at each pixel.
What is the intensity of light emitted by the object at the visible point?

Consider 16 pixels below, the visible object is at the right of the red line:



A pixel is not a dot, but a small square area.
If only one sample at the center of each square determines visibility, then lines in the rendered imaged appear jagged.

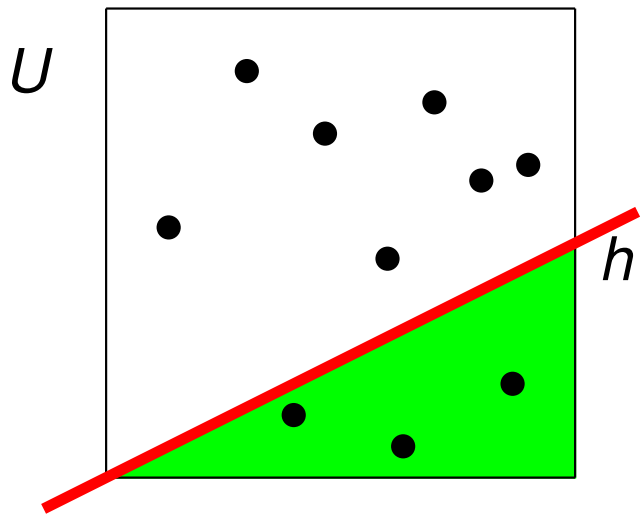
supersampling and discrepancy

More than one sample per pixel determines the visibiliy.

Definition (discrepancy of a sample set)

Let U be a square, S a sample set, and h the half plane which defines the visible area $h \cap U$. The area of $h \cap U$ is $\mu(h)$. The fraction of sample points in the visible area is $\mu_S(h) = \#(S \cap h)/\#S$.
The *discrepancy of h with respect to the sample set S* is

$$\Delta_S(h) = |\mu(h) - \mu_S(h)|.$$

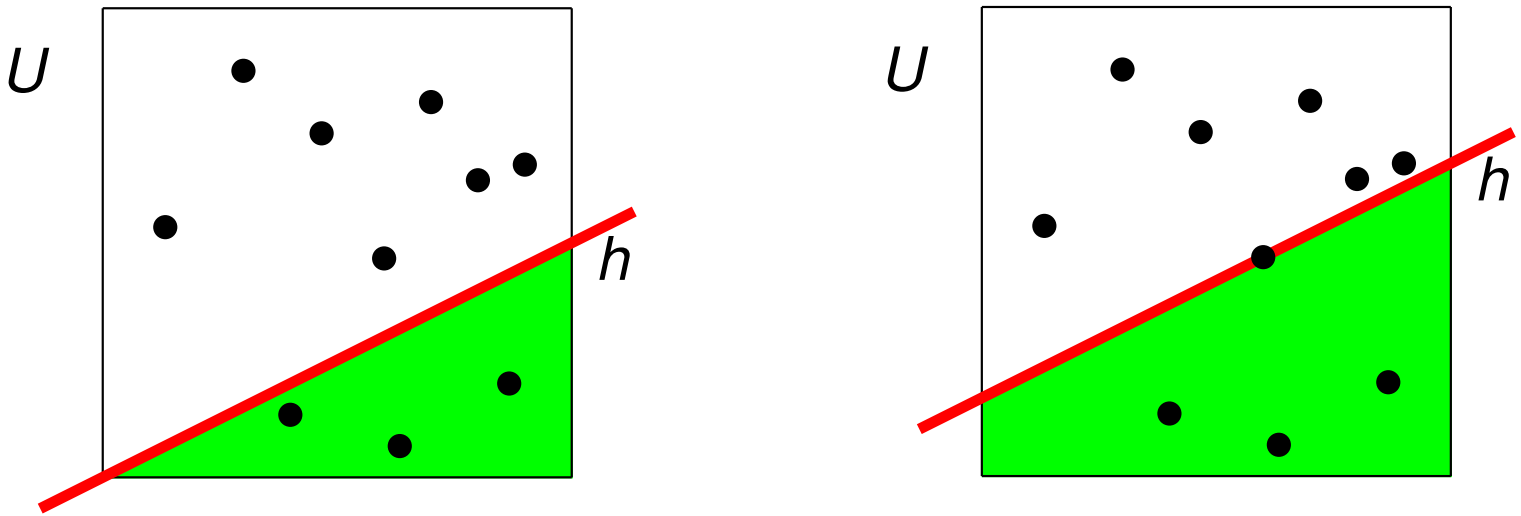


$$\#S = 10$$
$$\mu(h) = 1/4$$
$$\mu_S(h) = 3/10$$
$$\Delta_S(h) = |1/4 - 3/10| = 1/20$$

half plane discrepancy

Let $\mu(h) =$ area of $h \cap U$ and $\mu_S(h) = \#(S \cap U)/\#S$,
the discrepancy of h with respect to S is $\Delta_S(h) = |\mu(h) - \mu_S(h)|$.

The discrepancy increases as the half plane h increases:



Definition (half plane discrepancy)

The *half plane discrepancy of S* is the supremum of the discrepancies over the set $\mathcal{H}$ of all half planes: $\Delta_{\mathcal{H}} = \sup_{h \in \mathcal{H}} \Delta_S(h)$.

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working with a finite set of hyperplanes

The half plane discrepancy $\Delta_{\mathcal{H}} = \sup_{h \in \mathcal{H}} \Delta_S(h)$
is a supremum because the set $\mathcal{H}$ of all hyperplanes is not a finite set.

Suppose h contains no point of S on its boundary,

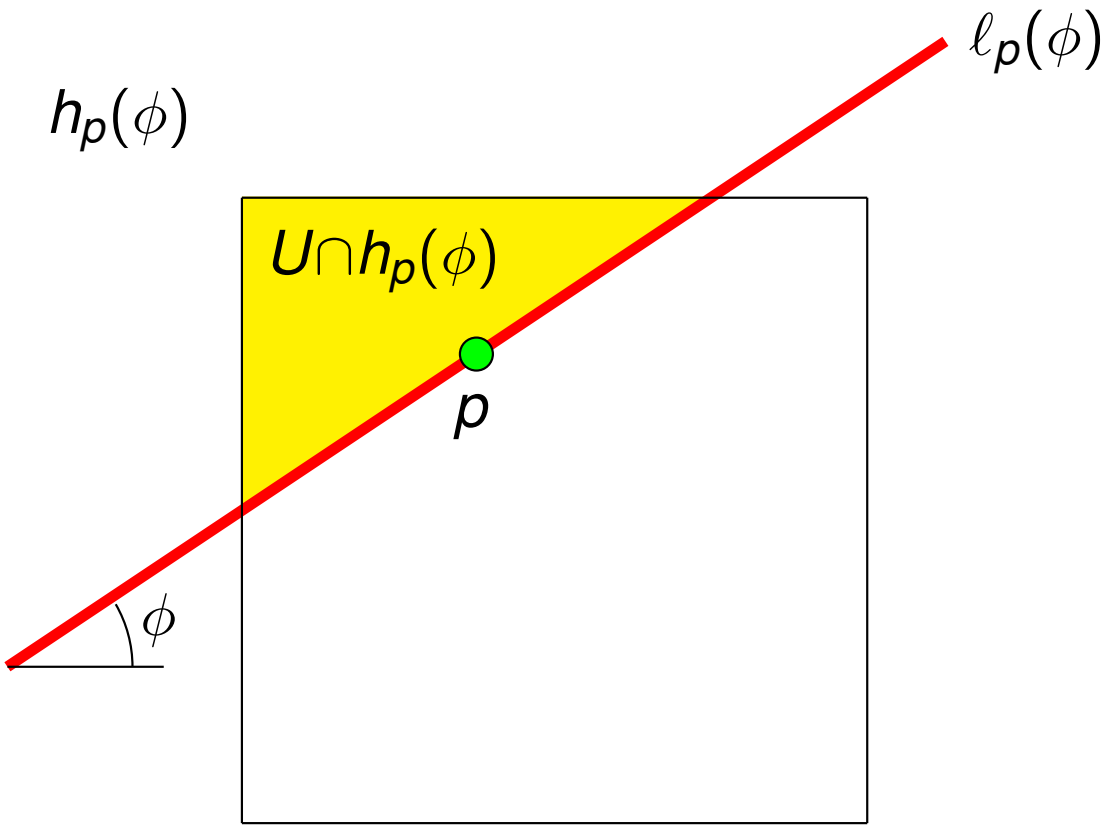
- shifting h changes $\mu(h)$, but not $\mu_S(h)$, as $\Delta_S(h) = |\mu(h) - \mu_S(h)|$,
- increasing $\Delta_S(h)$ in one direction,
  decreasing $\Delta_S(h)$ in another direction.

For $\Delta_S(h)$ to be an extremum,
the boundary of h must contain at least one point of S.

Does half plane with maximal discrepancy
always have two points of S on its boundary?

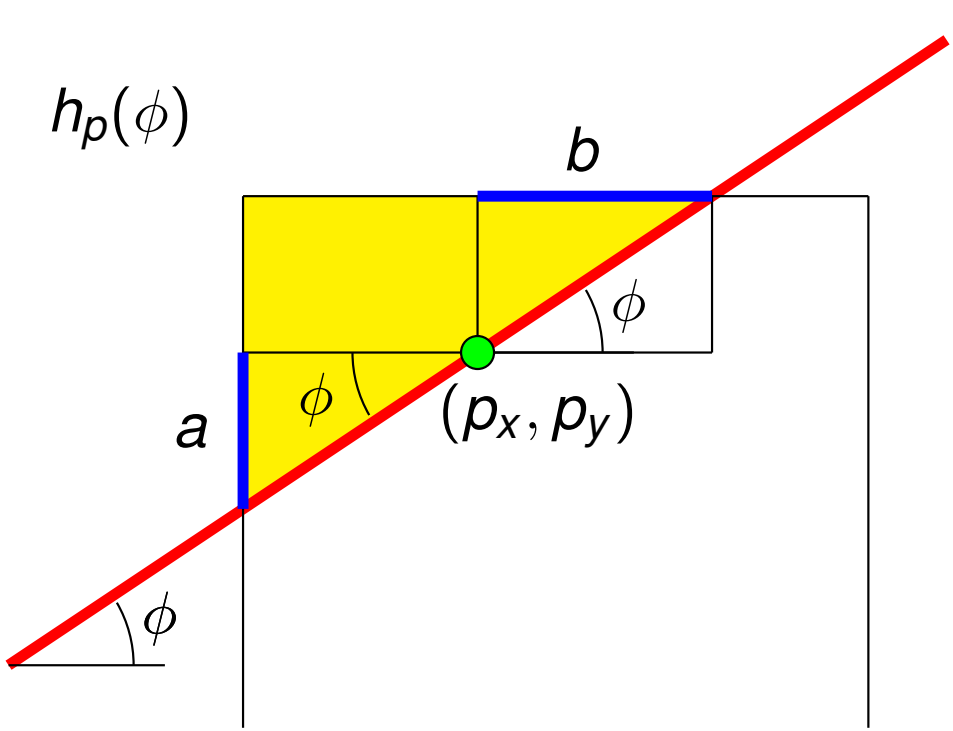
a line through a point

The line $\ell_p(\phi)$ through p is parametrized by the angle ϕ.



The half plane above $\ell_p(\phi)$ is $h_p(\theta)$.
Problem: maximize the area of $h_p(\theta) \cap U$, as a function of $\theta \in [0, 2\pi[$.

maximizing the area



$$\tan(\phi) = \frac{1-p_y}{b}$$

$$\tan(\phi) = \frac{a}{p_x}$$

$$b = \frac{1-p_y}{\tan(\phi)}$$

$$a = p_x \tan(\phi)$$

$$\begin{aligned}
\mu(h_p(\phi)) &= \frac{1}{2} p_x \left(p_x \tan(\phi) \right) + p_x \left(1 - p_y \right) + \frac{1}{2} \left(1 - p_y \right) \frac{1-p_y}{\tan(\phi)} \\
&= \frac{1}{2} \left(1 - p_y + p_x \tan(\phi) \right) \left(p_x + \frac{1-p_y}{\tan(\phi)} \right)
\end{aligned}$$

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the maximum discrepancy

Compute ϕ as the solution of

$$\max_{\phi \in [0,2\pi[} \Delta_S(h_p(\phi)) = \max_{\phi \in [0,2\pi[} \left| \mu(h_p(\phi)) - \mu_S(h_p(\phi)) \right|$$

where

$$h_p(\phi) = \frac{1}{2} \left(1 - p_y + p_x \tan(\phi) \right) \left(p_x + \frac{1 - p_y}{\tan(\phi)} \right)$$

and $\mu_S(h_p(\phi))$ is *the discrete measure*

$$\mu_S(h_p(\phi)) = \frac{\#(S \cap h_p(\phi))}{\#S}.$$

Two problems:

1. Determine the angle ϕ where the maximum is attained.
2. Compute the discrete measure, solving a counting problem.

the shape of $h_p(\phi)$

Exercise 1: Consider a point $p = (p_x, p_y)$ and

$$h_p(\phi) = \frac{1}{2}\left(1 - p_y + p_x \tan(\phi)\right)\left(p_x + \frac{1 - p_y}{\tan(\phi)}\right).$$

1. Plot $h_p(\phi)$ for some good values for p.
2. Describe its shape.

computing extrema

As ϕ goes from 0 to 2π, $\ell_p(\phi)$ may hit the vertices of U, the candidate extrema, at most eight times.

Lemma (types of extrema)

Let S be a set of n points in the unit square U.
A half plane h that achieves the maximum discrepancy with respect to S is of one of the following types:

1. *h contains one point $p \in S$ on its boundary.*
2. *h contains two or more points of S on its boundary.*

The number of the first type candidates is $O(n)$ and they can be found in $O(n)$ time.

The number of second type candidates is $O(n^2)$, to be dealt with later.

The first type of candidates can be computed in a straightforward manner in total $O(n^2)$ time.

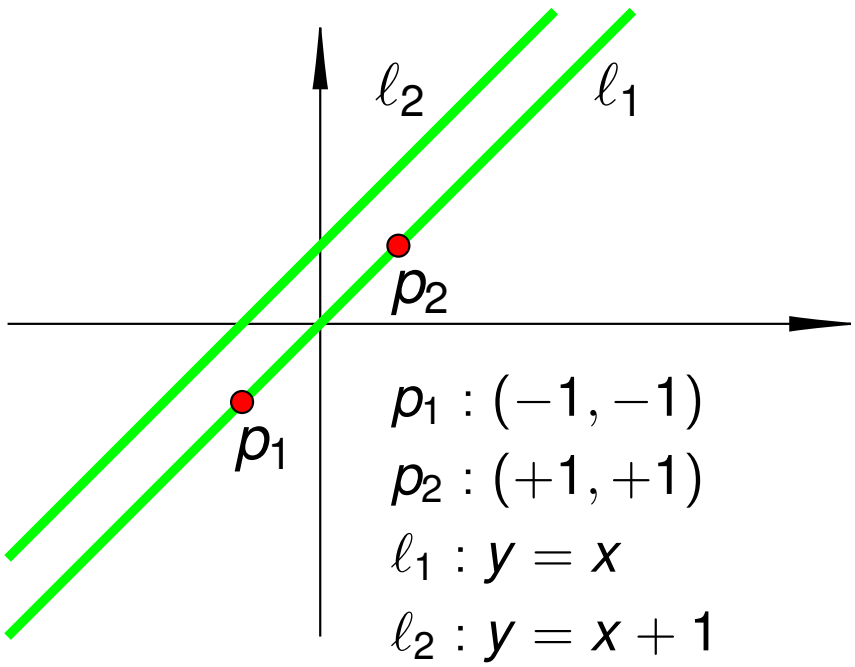
Discrepancy and Duality

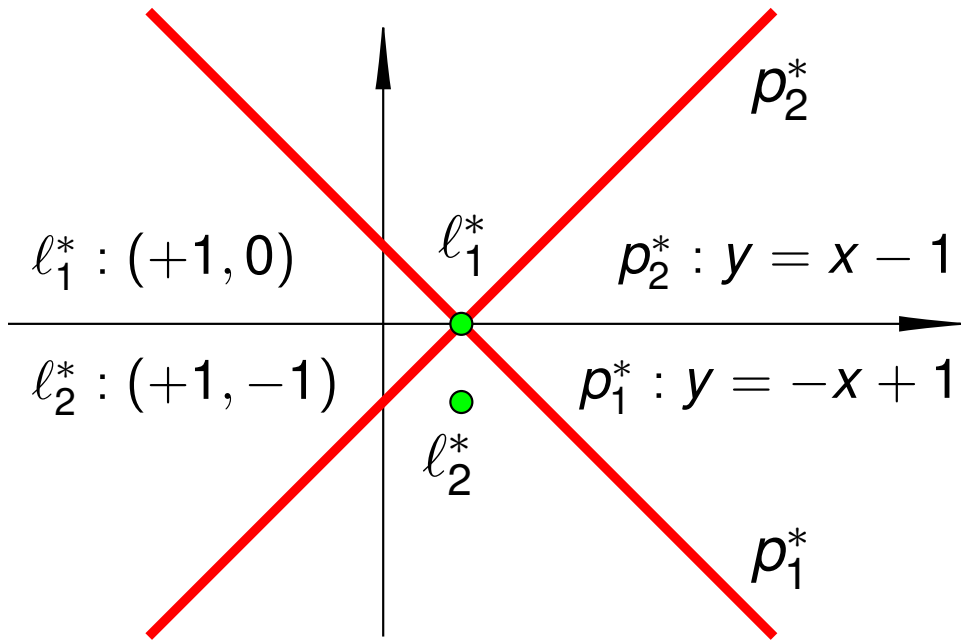
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dual lines and dual points

Definition (dual lines and dual points)

Given a point p with coordinates (p_x, p_y),
the dual p^* of the point p is the line $y = p_x x - p_y$.
Given the line ℓ with slope m and intercept b, $y = mx + b$,
the dual ℓ^* of the line ℓ is the point with coordinates $(m, -b)$.

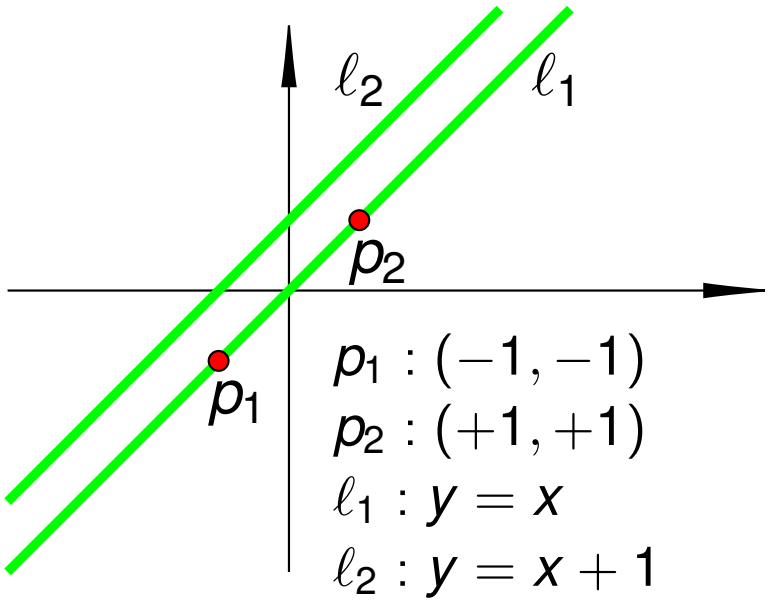


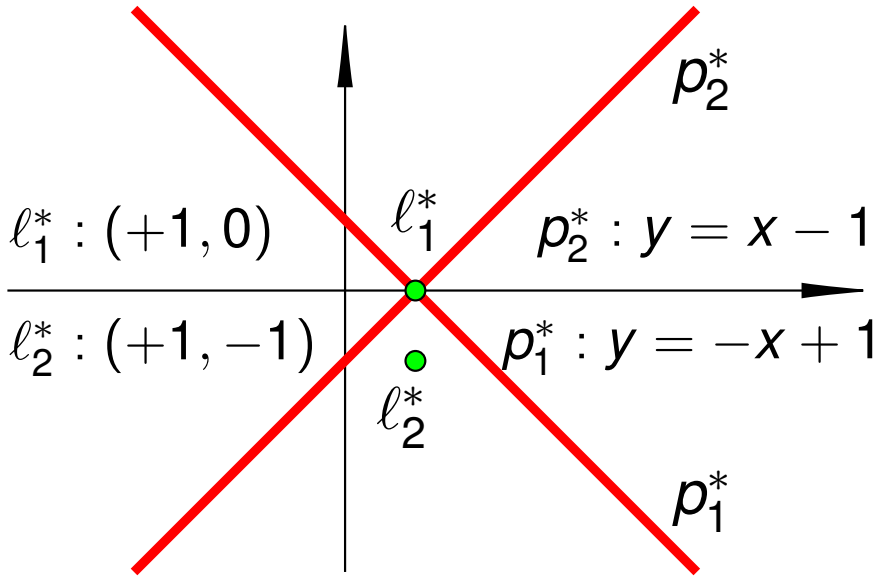


observations on duality

$p : (p_x, p_y)$, $\ell : y = mx + b$,

$p^* : y = p_x x - p_y$, $\ell^* : (m, -b)$





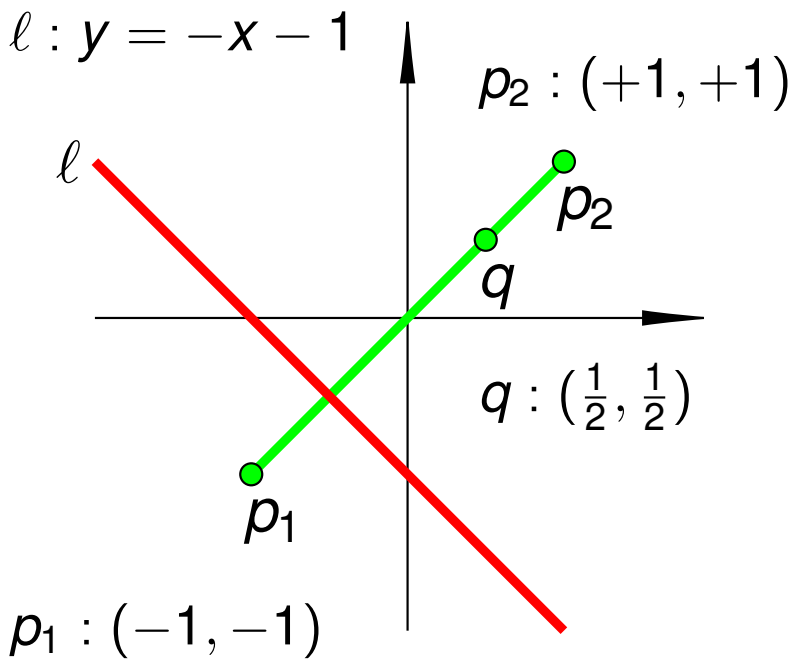
1. the dual of the dual is the primal: $(p^*)^* = p$, $(\ell^*)^* = \ell$
2. incidence preserving: $p \in \ell \Leftrightarrow \ell^* \in p^*$
3. order preserving: p above $\ell \Leftrightarrow \ell^*$ above p^*

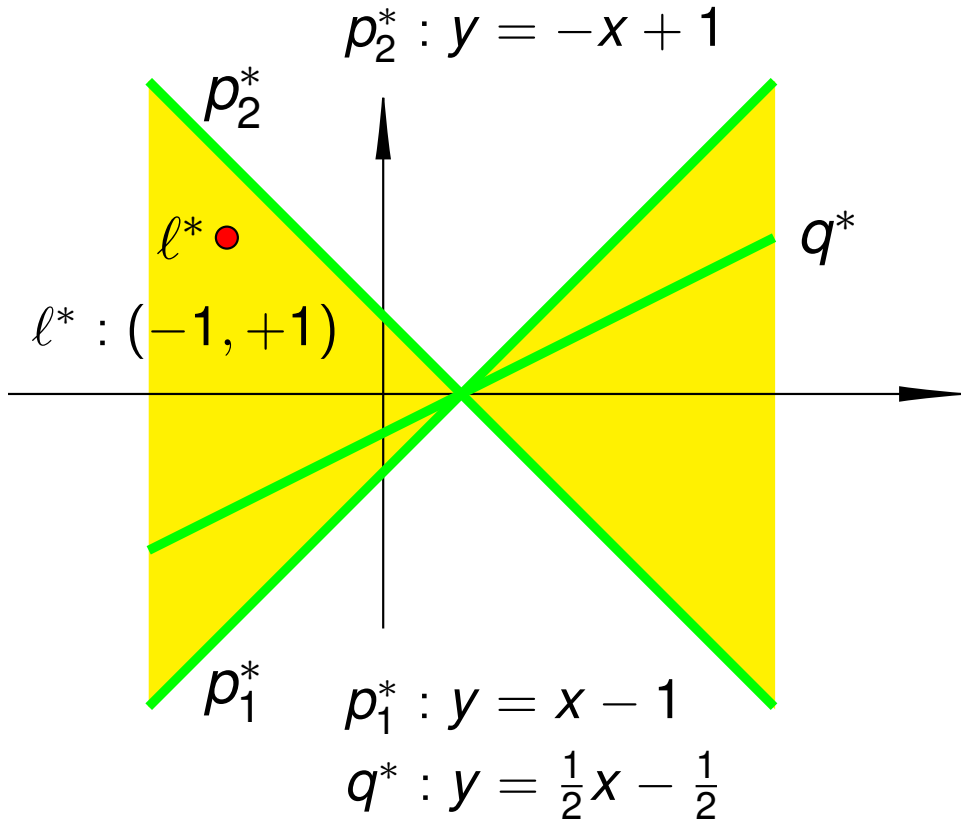
Exercise 2: Prove the above observations.

the dual of a line segment

$p : (p_x, p_y)$, $\ell : y = mx + b$, $\quad p^* : y = p_x x - p_y$, $\ell^* : (m, -b)$

Consider the line segment (p_1, p_2):





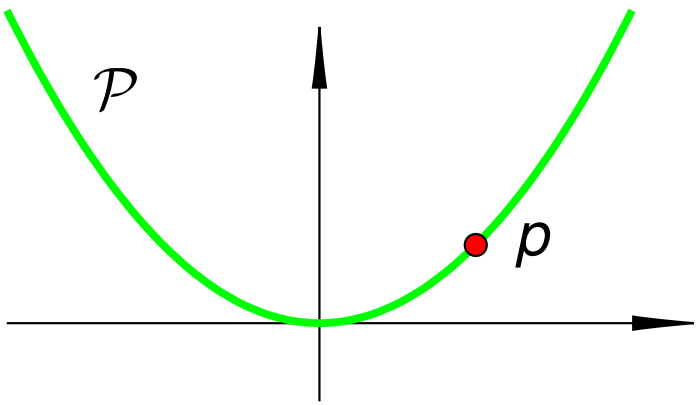
The dual of a line segment is a *double wedge*.

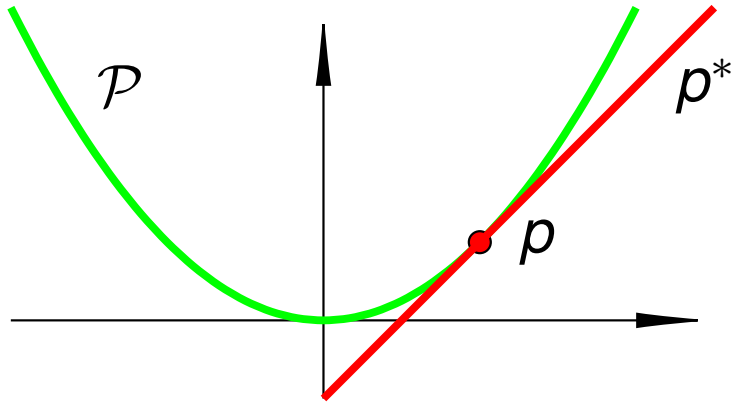
the dual transform on a parabola

$p : (p_x, p_y)$, $\ell : y = mx + b$, $\quad p^* : y = p_x x - p_y$, $\ell^* : (m, -b)$

Consider the parabola $\mathcal{P} : y = x^2/2$.

For every point $p \in \mathcal{P}$: $p = (p_x, p_x^2/2)$ and $p^* : y = p_x x - p_x^2/2$.





$p : (1, 1/2)$, $p^* : y = x - 1/2$ is the tangent line to $\mathcal{P}$ at p.

Exercise 3: Draw the dual of the parabola $y = x^2/2$.

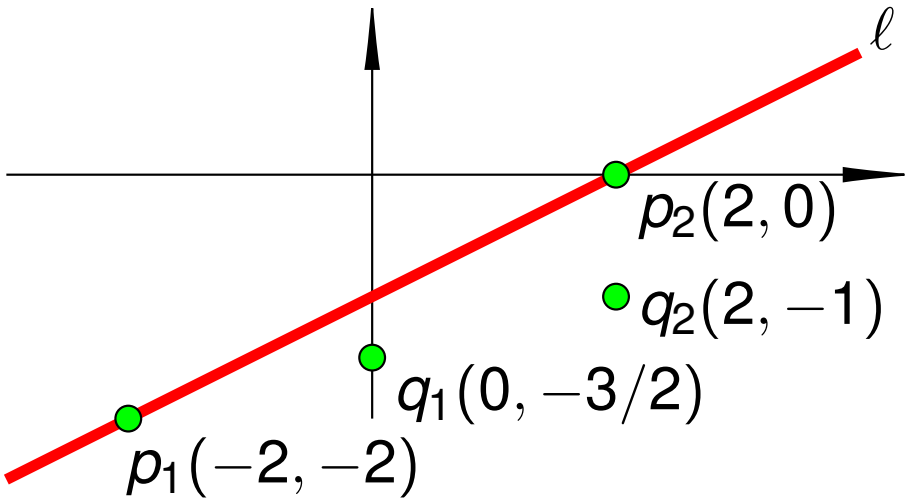
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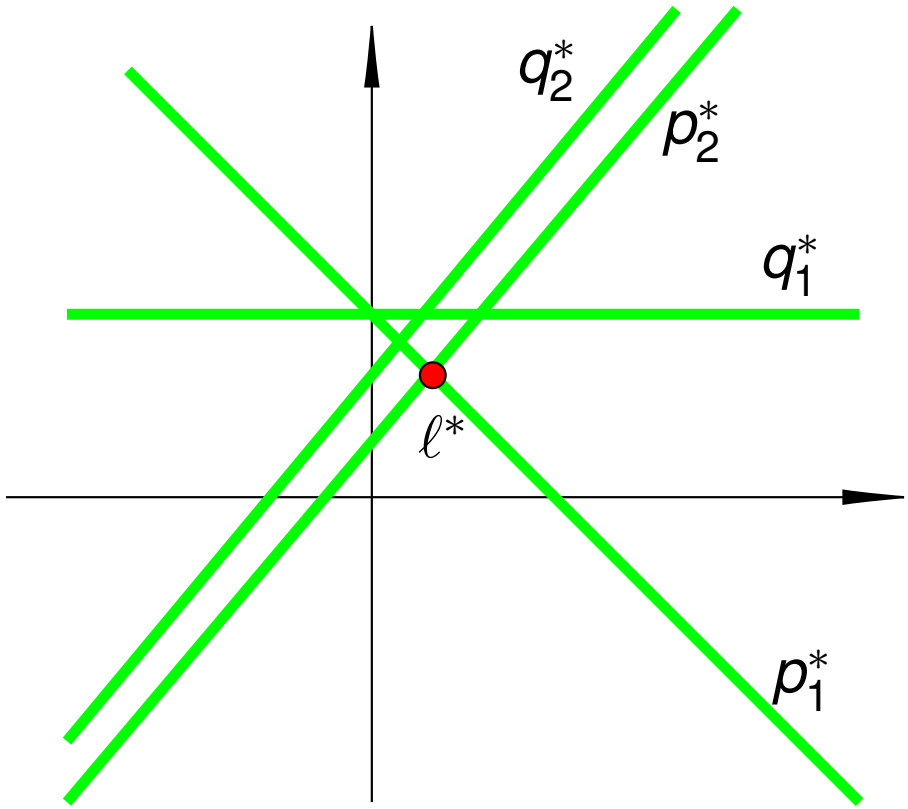
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the dual of the discrepancy problem

Given a line ℓ, we want to count all points below ℓ.

$\ell : y = (1/2)x - 1$
$\ell^* : (1/2, 1), q_1^* : y = 3/2$
$p_2^* : y = 2x, q_2^* : y = 2x + 1$
$p_1^* : y = -2x + 2$





Given the point ℓ^*, count the lines above ℓ^*.

counting lines

Given the point ℓ^*, count the lines below ℓ^*.

- Handle vertical lines separately, in time $O(n^2)$.
- For general lines, we consider arrangements of lines.

In the next two lectures:

1. an incremental algorithm to compute the subdivision defined by a set of lines will be described; and
2. the zone theorem on the complexity of an arrangement of lines will be proven.

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finding collinear points via duality

In `examples/Arrangements_on_surface_2` of the software CGAL, the program `dual_lines.cpp` illustrates an application of arrangements of lines.

Given a set of points, does the set contain three collinear points?

The points p_1, p_2, p_3 lie on the line ℓ

$\Leftrightarrow$ the point ℓ^* lies on the lines p_1^*, p_2^*, p_3^*.

The file `points.dat` contains 100 points, no collinear ones.

Exercise 4: Run the code on a point set with collinear points.

Combinatorial Geometry

The 1987 book *Algorithms in Combinatorial Geometry* by Herbert Edelsbrunner starts with arrangements as a fundamental concept.

Some combinatorial problems:

- count faces and incidences
- combinatorial equivalance
- encoding arrangements and configurations
- lower and upper bounds
- complexity of families of cells

Sylvester's Problem (1893)

Problem (Sylvester, 1893)

Let n given points have the property that the line joining any two of them passes through a third point of the set.

Must the n points all lie on one line?

The answer to this problem is yes,
known as the Sylvester-Gallai theorem.

For a short, geometric proof, see the paper
A Problem of Collinear Points, by H. S. M. Coxeter,
in *The American Mathematical Monthly* 55(1): 26–28, 1948.

summary and exercises

Section 8.1 and 8.2 in the textbook were covered.

Consider the following activities, listed below.

1. Write the solutions to exercises 1, 2, 3, and 4.
2. Consult the CGAL documentation and example code on arrangements of lines.
3. Consider the exercises 1, 2, 3 in the textbook.